

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Photoelectric Effect & Einstein's Equation	19 PYQs	Photoelectric
Moderate	De Broglie Hypothesis & Matter Waves	6 PYQs	Matter Waves

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Photoelectric Effect & Einstein's Equation	$E = h\nu = \phi_0 + K_{\text{max}}$, $K_{\text{max}} = eV_s = h(\nu - \nu_0) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$
De Broglie Hypothesis & Matter Waves	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$, $\lambda_{\text{electron}} \approx \frac{1.227}{\sqrt{V}} \text{ nm}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Photoelectric Effect & Einstein's Equation (Very High Yield — 19 PYQs)

Core Principle: Quantum nature of light, work function ϕ_0 , threshold frequency ν_0 , and stopping potential V_s dependence on frequency.

■■ **Caution / JEE Pitfall:** Stopping potential depends solely on FREQUENCY and work function; it is completely independent of intensity of incident light!

■ **Speed Trick:** Slope of stopping potential vs frequency curve is always universal: Slope = h/e .

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential)..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Work function of the metal is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The anode voltage of a photocell is kept fixed. The frequency of the light falling on the cathode is gradually increased. Then the correct g..." [Ans: N/A]

3.2 De Broglie Hypothesis & Matter Waves (Moderate Yield — 6 PYQs)

Core Principle: Dual wave-particle nature of moving particles, de Broglie wavelength under electrostatic acceleration V , and thermal gas particles.

■ **Caution / JEE Pitfall:** For neutral particles (neutrons), de Broglie wavelength is $\lambda = \frac{h}{\sqrt{3mkT}}$, where k is Boltzmann constant.

■ **Speed Trick:** For proton vs alpha particle accelerated through same voltage V : $(\lambda_p / \lambda_\alpha) = \sqrt{(m_\alpha q_\alpha / m_p q_p)} = \sqrt{(4 \times 2)} = 2\sqrt{2}$.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "If we consider an electron and a photon of same de-Broglie wavelength, then they will have same..." [Ans: N/A]
- **[2022 Shift] Q#3 (MCQ):** "The de-Broglie wavelength of a particle of kinetic energy K is λ , the wavelength of the particle, if its kinetic energy $\frac{K}{4}$..." [Ans: N/A]
- **[2020 Shift] Q#1 (MCQ):** "The de-Broglie wavelength associated with electron of hydrogen atom in this ground state is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential)

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: In the photoelectric effect, photoelectric current changes with anode potential in this way: 1. At zero anode potential, the current is still present. This happens because the emitted photoelectrons already have some kinetic energy, so some of them can reach the anode even without extra attraction. 2. When the anode is given a negative potential, it repels the photoelectrons. So, fewer electrons reach the anode and the current decreases. 3. At a particular negative potential, even the fastest photoelectrons cannot reach the anode. The current becomes zero at this point. This potential is called

Benchmark Problem 2: KCET 2025 (Q#3)

If we consider an electron and a photon of same de-Broglie wavelength, then they will have same

- (A) Angular momentum
- (B) Energy
- (C) Velocity
- (D) Momentum

Correct Answer: None

Solution Analysis: They'll have the same momentum. Reasoning (from de Broglie's relation): $\lambda = \frac{h}{p}$ $p = \frac{h}{\lambda}$. If an electron and a photon share the same λ , they share the same p .